



CAL-PLASMA SYMPOSIUM | FRESNO STATE

Modeling N_2/Ar Plasmas

Modeling of N_2/Ar Plasmas in somaFOAM; Project 4



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1 Explanation of Project

```
/*-----*- C++ -*-----*/
|=====|
| \ \ \ \ \ \ \ \ | F ield      | foam-extend: Open Source CFD
| \ \ \ \ \ \ \ \ | O peration | Version:      4.0
| \ \ \ \ \ \ \ \ | A nd        | Web:         http://www.foam-extend.org
| \ \ \ \ \ \ \ \ | M anipulation|
|-----*/
```

Objective

- Simulate an argon-nitrogen plasma using SOMAFOAM
- The simulation allows us to study plasma behavior without first performing expensive physical experiments.
- The main objective is to determine which operating conditions produce a stable and effective plasma.
- The project also investigates how simulation runtime can be reduced without significantly reducing accuracy.

Why Plasma Modeling is Important

- Low-pressure plasmas are used throughout semiconductor manufacturing for processes such as etching, deposition, cleaning, and surface treatment.
- Using simulations can help reduce development time, cost, and the number of physical experiments required
- These plasmas contain many different neutral, excited, and charged species that interact through complex chemical reactions.
- Computer modeling helps researchers understand these interactions and test different process conditions before developing a physical process recipe.
- Plasma can be a tool to solve many different problems and has many applications.

2 Governing Mechanics

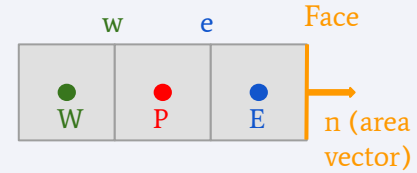
- Analyze mesh using the Finite Volume Method (FVM)
 - Φ is the transported quantity throughout the mesh (Phi, Temperature)
 - Each cell in the mesh has a centroid (node) where the Φ is located
- somaFOAM has schemes that are set in text files
 - Transient term (ddtSchemes) - Φ change with time

$$\partial(\rho\phi)/\partial t \approx \rho(\phi_p^{n+1} - \phi_p^n) / \Delta t$$

- Convection term (divSchemes) - Φ carried by flow
- Diffusion term (laplacianSchemes) - Φ spreading by gradients

- Why 1D?

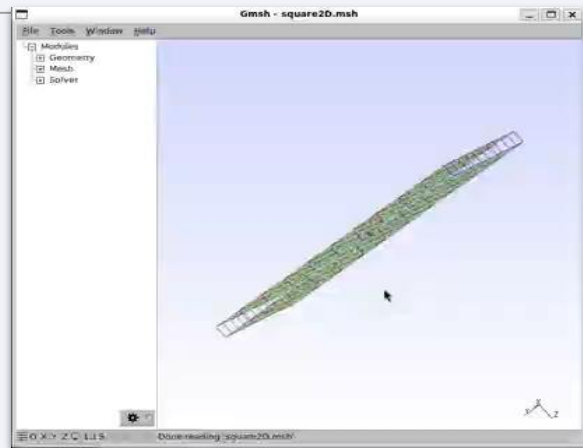
- Not as demanding (memory wise) compared to 2D & 3D simulations because of less complicated geometry and interactions
- Clearly demonstrates plasma behavior and can be used for global models



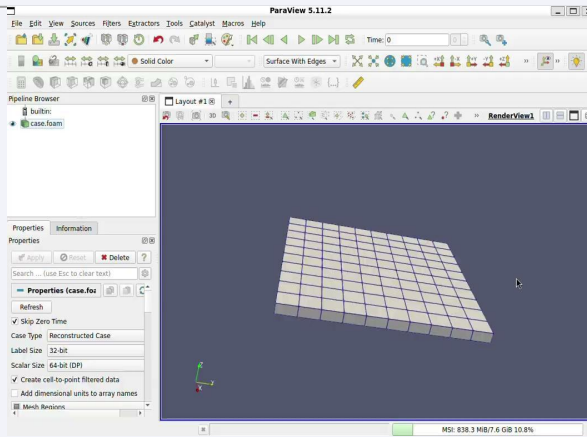
Convective Flux	Diffusive Conductance
$\downarrow$	$\downarrow$
$d/dx(\rho u \phi)$	$= d/dx(\Gamma d\phi/dx)$

2 Governing Mechanics

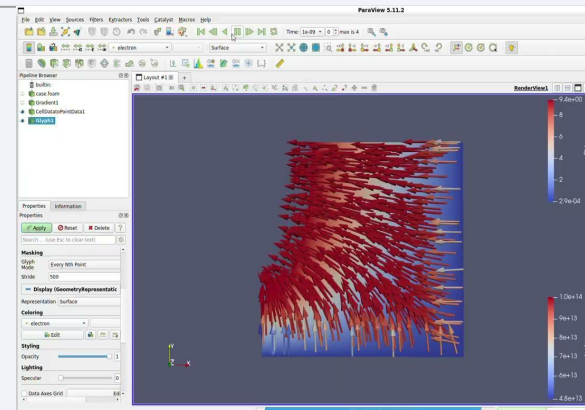
Gmsh: mesh created



ParaView: mesh visualization before solver



ParaView: solved result



Before solving the plasma case, the mesh must be created and verified. In each cell (397 cells total), the SOMAFOAM solver calculates values such as electric potential, temperature, and particle density. We can display the solved result in graphical or a video timelapse (as seen here).

Solved data that can be displayed in graphical or video form includes Phi, Electric potential/ voltage field, Arp1, Argon ion density, Electron, Electron density, Te, Electron temperature, E, Electric field, and more

3 Environment Information

Software/ Tool	Version	Description
Ubuntu Linux-Server	18.04.6 LTS, WSL@	Remote operating environment used to store cases, execute somaFOAM, and run mesh and solver utilities
SSH/ Remote Server Access		Provides remote terminal access to the school server and the shared project directory at /DATA/plasmaD
somaFOAM-Server	foam-extend 4.0; linux64GccDPOpt	Main plasma-modeling environment containing the N ₂ /Ar chemistry, field equations, and supporting solver libraries.
somaFoam- Solver	foam-extend 4.0 build	somaFOAM solver; runs the plasma simulation and calculates species densities, electric potential, electron temperature, and other fields over a determined time.
Gmsh- Local	4.12.1	Creates and modifies the reactor geometry and mesh, including the electrode gap and cell refinement near the sheath regions.
gmshToFoam- Server	foam-extend 4.0 utility	Converts the Gmsh .msh file into the constant/polyMesh format required by somaFOAM.
checkMesh-Server	foam-extend 4.0 utility	Checks mesh dimensions and qualities; boundary patches, cell quality, face areas, whether the mesh is correctly configured (1D/2D/3D) and more.
ParaView- Local	5.11.2	For visualizing the mesh and simulation results changing over time; including Phi, electric field, electron temperature, species densities, line plots and more.
Case-Server	Various	Provides the baseline geometry, initial fields (/0), boundary conditions, reaction chemistry, solver settings and more.

Workflow using tools: Connect to server → Navigate to case → Activate somaFOAM → Edit case files → Generate mesh with Gmsh, if not given → Convert with gmshToFoam → Validate with checkMesh → Run somaFoam → Save and monitor solver logs → Analyze data in ParaView

Stage 1:

1 Understand the N_2/Ar Model

- Learn how to navigate cases and adjust their chemistries

- Understand case contents: break down the predefined somaFOAM cases to understand the plasma species, initial number densities, boundary conditions, reaction chemistry, transport properties, and solver settings

- Read papers & watched tutorials to understand how to use practice examples

Stage 2:

2 Practice cases

- Configure environment to be usable for needed softwares and understand how to use the softwares

- Run practice examples of different powers for learning how to navigate environment and softwares

- Learn how to troubleshoot the solver and output errors

- Learn how to analyze the mesh, electric potential, electric field, temperatures, and species-density results in ParaView

Stage 3:

3 Run control case

- Access server for solves with bigger load

- Successfully run the control case and get baseline data

- Understand our baseline data

Stage 4:

4 Data Analysis

- Adjusting parameters of completed plasma cases in order to determine optimal conditions for a plasma jet

- Compare data to previous example cases and future control cases in order to further optimize parameters

Where we are now: The control case has been partially completed and its baseline data is ready; we are now validating the results and transitioning from Stage 3 into the comparative data analysis of Stage 4.

Set Up (geometry, mesh, boundary conditions)

- Begin with **N₂/Ar-30** somaFOAM case with predefined chemistries- our control case
- Review contents of the case. Case structure:
 - **0/** — initial species, temperatures, potential, and boundary conditions
 - **constant/** — plasma chemistry, reaction data, transport properties, and mesh
 - **system/** — solver controls, time-step settings, and output instructions

- Adjust chemistry (constant/)
 - Check reaction-rate values for settings that may cause rapid growth, instability, or numerical errors. These values may need adjustment when errors are encountered during solver runs.
- Review the plasmaProperties settings

Analyze Mesh (add file path)

- Run checkMesh to verify mesh quality and confirm dimensions.
- Modify the original Gmsh geometry and regenerate the OpenFOAM mesh when the gap or cell distribution is changed based on solver and simulation needs. Reducing the gap, adjusting number of cells, adjusting power, changing 1D/2D/3D.

Run Solver (somaFoam)

- Monitor the solver output to confirm the simulation remains stable and identify, diagnose, and correct errors as they occur.

Analyze Data (quantities, extracted, comparison made)

- Use ParaView to examine:

◦ Electric potential, Phi	◦ Electron and positive-ion number densities
◦ Electric field, E_X	◦ Identify the bulk plasma, sheath regions, and conditions where the plasma becomes weak and unstable.
◦ Average electric field, EMean_X	◦ Compare the data of different simulations
◦ Electron temperature, Te	

5 Control Case Information

N₂/Ar Control Case Configuration

Control-Case Conditions

- 80% Ar / 20% N₂
- 1 atm
- 300 K
- 2 mm electrode gap
- 1D model

The result of checking the mesh shows that the modeled electrode gap is 2 mm, extending from $x = 0$ to $x = 0.002$ m. It shows mesh size, cell count, and cell type. The mesh divides the 2 mm space between the electrodes into 397 small cells so the solver can calculate how the plasma changes across the gap.

Mesh OK.

```
Mesh stats
all points:      1592
all faces:      1986
internal faces:  396
cells:          397
boundary patches: 6
Overall number of cells of each type:
hexahedra:      397
```

\$ checkMesh

```
Checking geometry...
This is a 1-D mesh
Overall domain bounding box (0 0 -0.001) (0.002 0.001 0)
Mesh (non-empty, non-wedge) directions (1 0 0)
Mesh (non-empty) directions (1 0 0)
Mesh (non-empty, non-wedge) dimensions 1
```

constant/electroMagnetics

```
mode      continuousFrequencyModulated;
initialAmplitude 200;
frequency  13.56E6;
bias       0;
power      0.0955;
```

This file controls the electrical power applied to the plasma. The source operates at 13.56 MHz, begins at 200 V, and uses a power setting of 0.0955.

system/controlDict

```
application      plasmaSimFoam;
startFrom        latestTime;
startTime        0;
stopAt           endTime;
endTime          5E-6;
writeInterval     3.6873156E-9; // T/20 -> 20 writes per RF cycle (13.56 MHz)
```

This file controls how the simulation runs. It tells the solver where to restart, how long to run, the size of each time step, and how often results are saved

0/Ar

```
internalField      uniform 1.9570506334E25;
// 1 atm, 300 K, 80% (was 2.575059507038598E22)
```

The density was changed to 80% Argon and 20% nitrogen

0/N2

```
internalField      uniform 4.8926265836E24;
// 1 atm, 300 K, 20% (was 6.437648767596495E21)
```

constant/polyMesh/boundary

```
front      type: empty
top        type: empty
leftelectrode type: patch
bottom     type: empty
back       type: empty
rightelectrode type: patch
```

The mesh is 1D because the simulation only models changes along the line between the left and right electrodes; the other directions are excluded (marked empty)

constant/plasmaProperties

```
diffusionModel  einsteinRelation;
mobilityModel   eTemp;
collisionFrequency muBased;
transportModel  driftDiffusion;
```

These settings control how electrons and ions move through the gas

0/Phi

```
internalField      uniform 200;
```

This sets the starting electric potential through the plasma region to 200v. The electrode boundary conditions then control the voltage at each side during the simulation.

5 Control Case Information

Neutral Species	<code>uniform 1.9570506334E25; // 1 atm, 300 K, 80%</code> <code>uniform 4.8926265836E24; // 1 atm, 300 K, 20%</code>	Ar - 80%, N2 - 20%
Pressure	<code>uniform 101325; // 1 atm</code>	The pressure on the system is 1 atm (101325 Pa)
Temperature	<code>uniform 300;</code>	The temperature of the plasma is 300 Kelvin
Mesh size (electrode gap)	<code>Overall domain bounding box</code> <code>(0 0 -0.001) (0.002 0.001 0)</code>	Length between the left and right electrode is 0.002m (2mm)
Model Type	<code>\$ checkMesh</code> <code>Checking geometry...</code> <code>This is a 1-D mesh</code>	The checkMesh function verifies mesh quality and determines that the left and right faces are patches, while the others are empty
Electric Potential	<code>internalField uniform 200;</code>	The starting electric potential is 200V
Frequency	<code>frequency 13.56E6;</code>	The source operates at 13.56MHz
Timestep (deltaT)	<code>Solver timestep Δt: 5E-11 s</code>	The time step is how far forward the simulation moves in time during each calculation. A smaller timestep gives more detail but takes longer.
Final simulation time	<code>1.78466072828081e-06</code>	The simulation models 1.78 microseconds of plasma behavior (supposed to be 5 microseconds)
Write interval	<code>writeInterval 3.6873156E-9;</code> <code>// T/20 -> 20 writes per RF cycle (13.56 MHz)</code>	The simulation saves data every 3.687 ns

Current State of Control Case

- Solver is running with 0 reported solver errors
- Total current run time is 36 hours 28 minutes
- 487 of approximately 1,358 total saved simulation time folders have been successfully saved
- Current simulated time is 1.77×10^{-6} (1.77 microseconds) and our target simulated time is 5.0×10^{-6} (5 microseconds)

- Solver runtime is a huge issue and must be addressed before running additional cases.
- Progress has been inconsistent and recently slowed to only 0.96% over 11 hours.
- It could take 70.5 hours total at the overall average rate, but 31 more days if the solver follows the recent slow rate.

5 Control Case Information

To reduce the solver time:

- We have already managed to cut down on run time by splitting the load by simultaneously running independent post processing programs to calculate derived fields of dependent data (such as means) while the main solver continues.
- Chemistry and solver settings have already been safely adjusted to improve speed but preserve accuracy of the data.

Further work is needed to reduce solve time and improve our data:

- **Reducing the power**
- Increase speed by using **mkfs.ext4** virtual drive because we right now are using something native to windows and not linux.
- The solver is primarily using one CPU core, despite the server having many additional cores available. Perhaps there are ways to use our environment more efficiently to reduce solving time. We should look into **parallel processing**

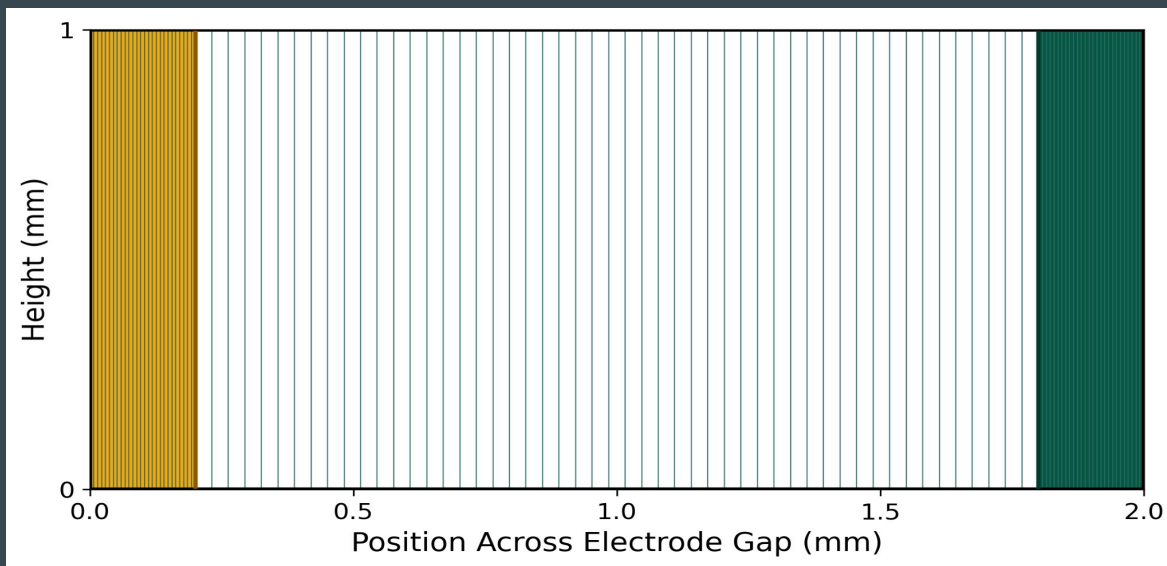
Aspects that could potentially reduce solver time, but could compromise data:

- Increasing timestep (deltaT) could reduce runtime, because smaller timesteps require more calculations to reach physical time. But this could impact accuracy of data.
- Reducing the final simulation time (endTime) makes the run finish sooner, but produces a short simulation.
- More cells means every plasma field and equation must be solved at more locations per timestep. Lowering mesh cell count could decrease run time but lose critical detail.
- Runtime increases with the number of reactions, equations, and calculations. We currently are working with 46 reactions.
- Reducing output frequency (writeInterval) could reduce the post processing work as so much data is written, but it does not reduce number of actual solver timesteps

6 Results

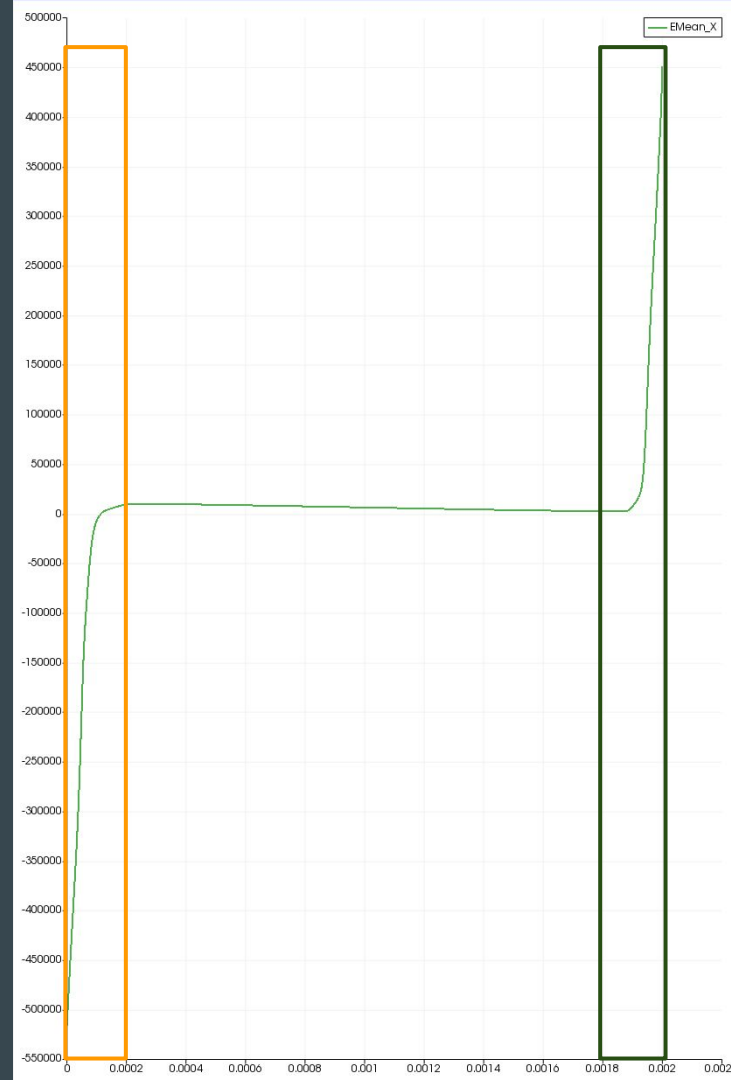
ArN2-30: Mesh

- The mesh below is $0.002\text{ m} \times 0.001\text{ m} \times 0.001\text{ m}$.
- The estimated sheath regions are located near the left and right electrodes, shown in orange and dark green.
- The preliminary sheath regions are approximately 0.0002 m long on each side.
- These edge regions contain the strongest electric-field gradients and therefore require greater mesh resolution.



ArN2-30: E_XMean - Electric Field

- Left sheathe region:
 - Highly negative ($\approx -500,000$ V/m)
 - A negative E_X means the electric field is pointing in the -x direction (to the left)
 - Ions that are accelerated to the left electrode will not strike with as much energy compared to the right electrode
- Right sheathe region:
 - Highly positive ($\approx 450,000$ V/m)
 - A positive E_X means the electric field is pointing in the x direction (to the right)
 - Ions are accelerated at much faster speeds towards the right electrode compared to the left electrode
- Bulk Region (quasi-neutral):
 - There is a low electric field in this region, essential for electrons and ions to drift and interact (≈ 7600 V/m)



6 Results

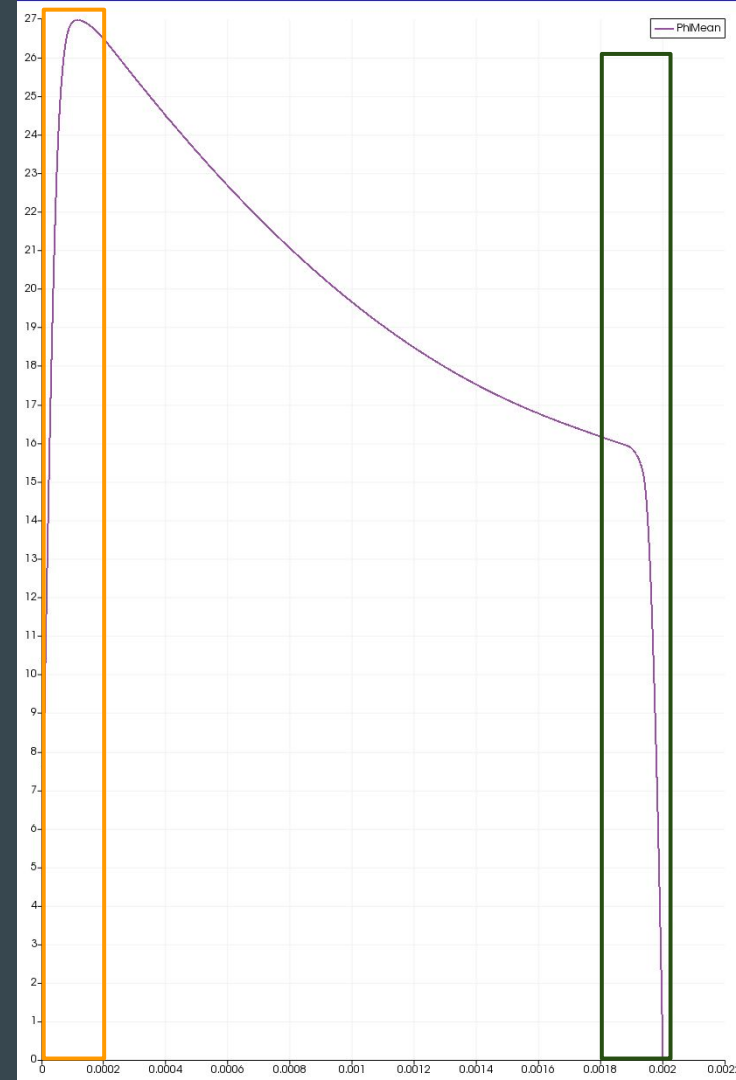
ArN2-30: Phi - Electric Potential

- Left sheathe region:
 - Peaks at $\approx 27V$, and drops down to $26.5V$ at the wall
- Right sheathe region:
 - Has a steep drop from $16V$ down to $0V$, and this can be related to the electric field
- Bulk Region:
 - The bulk region drops from $\approx 26V$ down to $\approx 16V$, which can be calculated below by using the electric field and length of mesh

$$\Delta V = E_x * m \quad \Delta V = \approx 7600 \text{ V/m} * 0.0016\text{m} =$$

12.16V

Note: This graph is a time-averaged, meaning the smaller voltage values are due to voltage swinging from $+200V$ to $-200V$



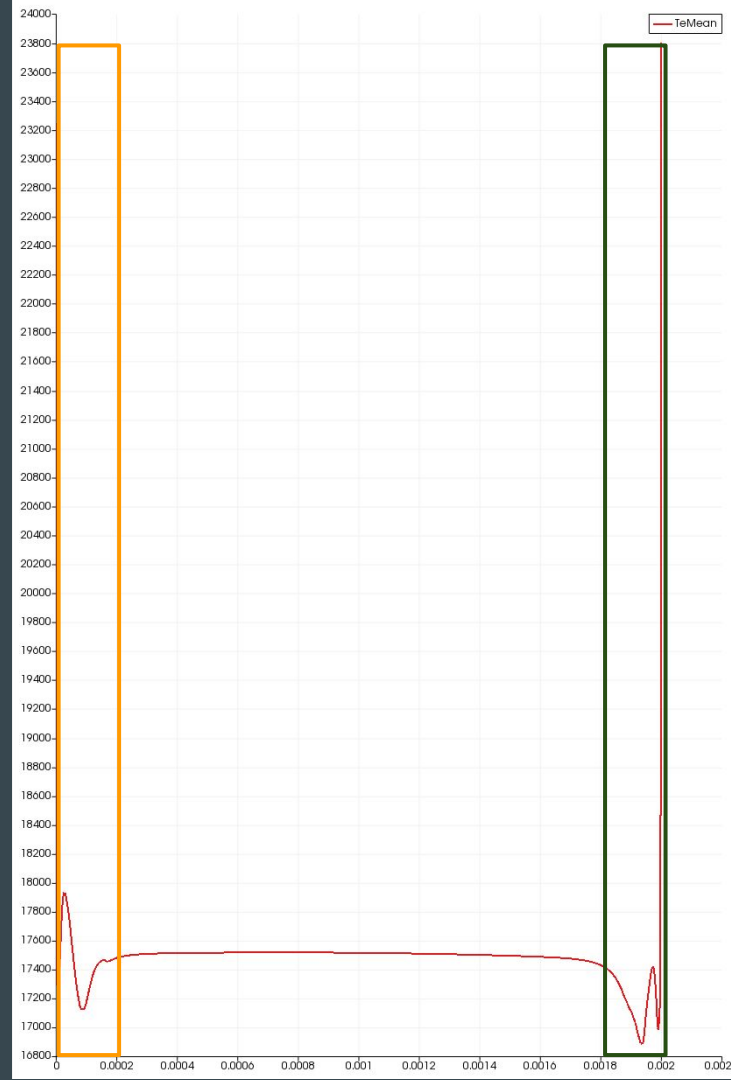
6 Results

ArN2-30: T - Temperature Te - Temperature of Electrons

T: The temperature of the left and right electrodes are set to 300K, meaning that we assume the gas will not change in temperature.

However, the initial temperature for Te is 10000K, and can change throughout the plasma reaching a bulk temp of roughly 17500K.

- The negatively charged particles are constantly accelerated in the bulk region
 - There are more electrons present in this region, meaning the temperature of the electrons is due to some of the electrons being accelerated back into the bulk from the sheathe region
- When electrons push past the initial sheathe region, the electrons tend to move faster towards the walls due to the high electric field
 - The electron temperature then rises since they are moving at a higher speed towards the wall

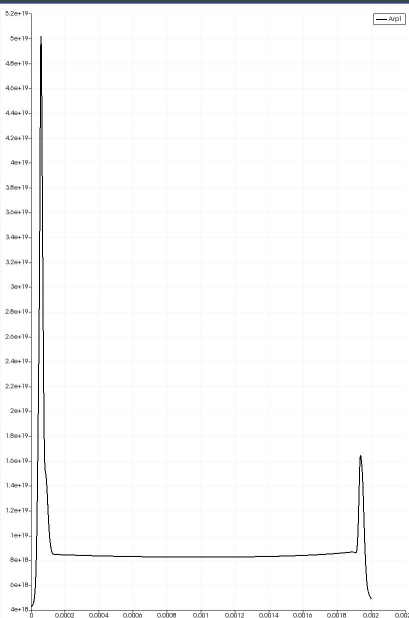


6 Results

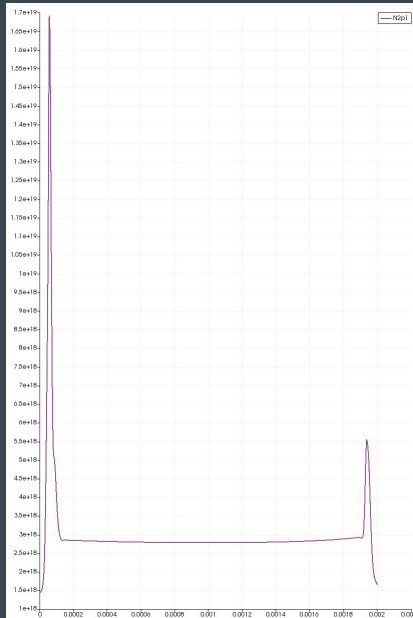
ArN2-30: Ion Density (Arp1, Np1, N2p1)

This section clearly shows that our solver encountered an issue

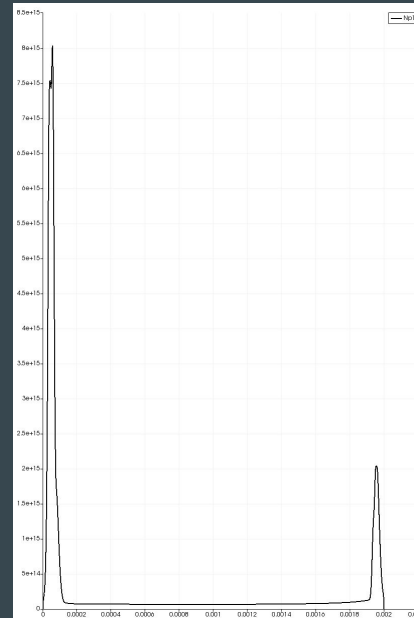
Ar Ion



N2 Ion



N Ion



- The left sheathe region experiencing a spike in number density for all ions present
 - Sheathe regions contain less electrons, however ion density is higher in the bulk region usually
- Possible suspects for this divide by zero error
 - Power: 0.0955W (per 1 micrometer²)

power 0.0955;

- Mesh size: mesh was originally created for 5mm instead of 2mm

Where this goes from here

1

Immediate

Continue working and optimizing the ArN2-30 high-pressure cold plasma case. Reduce run time, lower power, and use mkfs.ext4 virtual drive.

2

Rest of program

Continue modeling ArN2-cases, modifying parameters in order to create the best simulation for a high-pressure cold plasma jet

3

Beyond

Build a plasma jet to test how simulation compares to physical data

Thank you

Questions welcome



Group D · Faculty Mentor: Jose Alfredo Millan Higuera · July 28, 2026

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